If $y = b^x$, then $y' = b^x \ln b$

In general, if $y = b^{g(x)}$, then $y' = b^{g(x)} \cdot g'(x) \ln b$

Example: Differentiate the following.

a) $y = 4^x$

$y' = 4^x \cdot \ln 4$

$y' = (\ln 4) 4^x$

b) $y = 3^{x^2-4x}$

$y' = 3^{x^2-4x} \cdot (2x-4) \cdot \ln 3$

$y' = 2(x-2)(\ln 3) \cdot 3^{x^2-4x}$

c) $y = 5^{x^3+1} \cdot \sin(e^x)$

$y' = \underbrace{5^{x^3+1} \cdot (3x^2)(\ln 5) \sin(e^x)}_{1} + \underbrace{5^{x^3+1} \cdot \cos(e^x) \cdot e^x}_{2}$

$y' = 5^{x^3+1}(3x^2(\ln 5)\sin(e^x) + e^x \cdot \cos(e^x))$

$y = e^x \cdot \cos x$

poi

$\frac{3}{3}$

OMIT

Derivatives of Logarithms

$b = e$ (2)

$\ln g(x)$

If $y = \log_b x$, then $y' = \dfrac{1}{x \cdot \ln b}$

Proof: $y = \log_b x$

$$y = \frac{\log_e x}{\log_e b}$$

$$y = \frac{\ln x}{\ln b}$$

$$\therefore y' = \frac{1}{x \cdot \ln b}$$

If $y = \log_b g(x)$, then $y' = \dfrac{g'(x)}{g(x) \cdot \ln b}$

Example 1: Differentiate the following.

① $y = \log_3 x$ ② $y = \log_4 (x+1)$ ③ $\log_{\frac{e}{2}} x$

$$y' = \frac{1}{x \cdot \ln 3} \qquad y' = \frac{1}{(x+1) \cdot \ln 4} \qquad y' = \frac{1}{x \cdot \ln\left(\frac{e}{2}\right)}$$

$$y' = \frac{1}{x\left[\ln e - \ln 2\right]}$$

$$y' = \frac{1}{x\left[1 - \ln 2\right]}$$

4. $y = \log_2 [5x^2 - 11x]$

$$y' = \frac{10x - 11}{(5x^2 - 11x) \cdot \ln 2}$$

5. Find the equation of the tangent line to the curve $y = \log_5 (x^2 - 3x)$ when $x = 4$.

Solution:

$$\frac{dy}{dx} = \frac{2x-3}{(x^2-3x) \cdot \ln 5}$$

$$\left.\frac{dy}{dx}\right|_{x=4} = \frac{5}{4 \ln 5}$$

$$y - y_1 = m(x - x_1)$$

$$y - \log_5 4 = \frac{5}{4 \ln 5}(x - 4)$$

$$y - \frac{\ln 4}{\ln 5} = \frac{5}{4 \ln 5}(x - 4)$$

log

6. $y = \sin(x^2) \cdot \log_3(\sin(x))$

$$y' = \cos(x^2)(2x)\log_3(\sin(x)) + \sin(x^2) \cdot \frac{\cos(x)}{\sin(x) \cdot \ln 3}$$

$$y' = 2x\cos(x^2)\log_3(\sin(x)) + \frac{\cot(x)\sin(x^2)}{\ln 3}$$

$$y' = \frac{2x\cos(x^2)\ln(\sin(x)) + \cot(x)\sin(x^2)}{\ln 3}$$

7. Determine the equation of the normal line to the curve $y = \log_3(x^3 - 2x)$ at $x = 2$.

Solution

$$\frac{dy}{dx} = \frac{3x^2 - 2}{(x^3 - 2x) \cdot \ln 3}$$

$$\left.\frac{dy}{dx}\right|_{x=2} = \frac{10}{4\ln 3} \Rightarrow m_\perp = \frac{-4\ln 3}{10} = -\frac{2\ln 3}{5}$$

$$y - \log_3 4 = -\frac{2\ln 3}{5}(x - 2)$$

$$y - \frac{\ln 4}{\ln 3} = -\frac{2\ln 3}{5}(x - 2)$$